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GPU Architecture

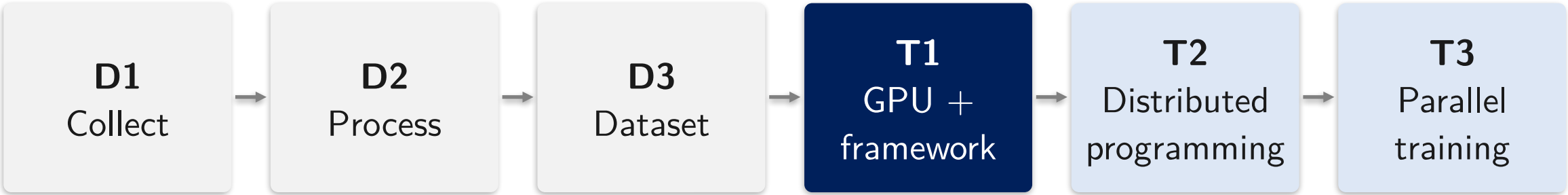
Why GPUs are fast, where they are slow, and how to predict which one you will get

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COMP 329 · Spring 2027 · Week 8, Lecture 1

Where we are in the pipeline

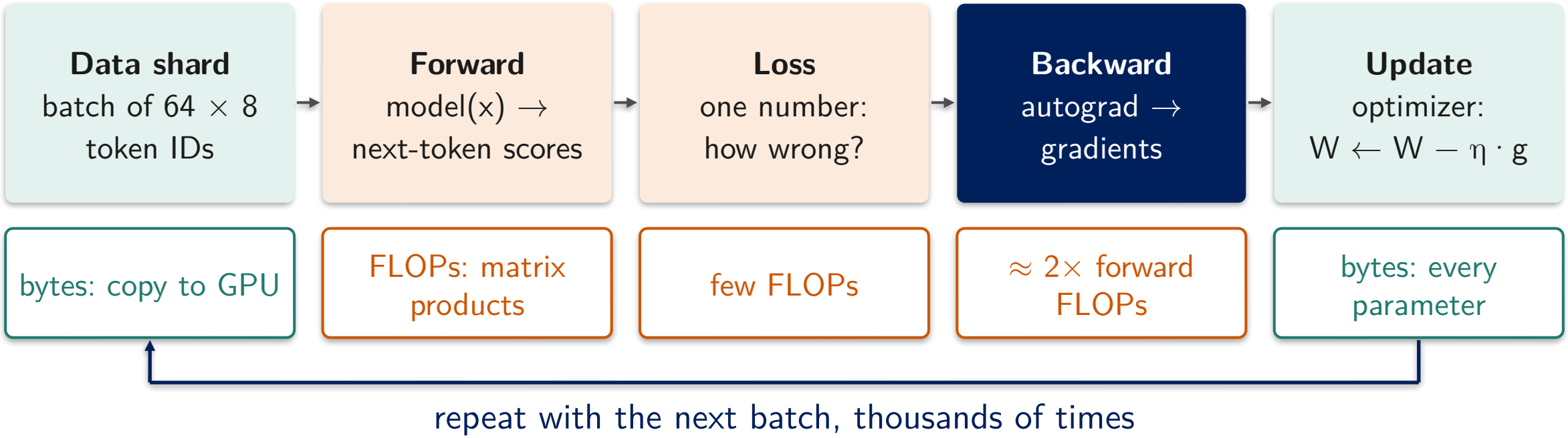


In our pipeline

Today: what happens inside **one GPU** running **one kernel** — the foundation for T1 and Lab 3.

Next: what one training step does, and the systems this block builds to make it fast.

What one training step does



Key idea

Training time = number of steps × time per step. The training block makes **one step fast**, then splits it across **many GPUs**.

Training systems at a glance

Problem	System component	Where
One step is millions of multiply-adds	**GPUs and kernels**	Wk 8 · T1
A kernel per op is impractical	**Frameworks and compilers**	Wk 9–10 · T1
One GPU is too slow or too small	**Processes + communication: MPI, NCCL**	Wk 10–11 · T2
How do we split the work?	**Data, tensor, pipeline parallelism**	Wk 12–13 · T3
Things break at scale	**Diagnosis**	Wk 13

dark: today

Why do we need GPUs?

The job: train GPT-3 (175 billion parameters)
$\approx$ **3.14 $\times$ 10^{23}** floating-point operations, almost all multiply-adds

If one CPU core did all of this work, how long would it take?

One **CPU core**, at its peak
$\approx$ **52,000 years**

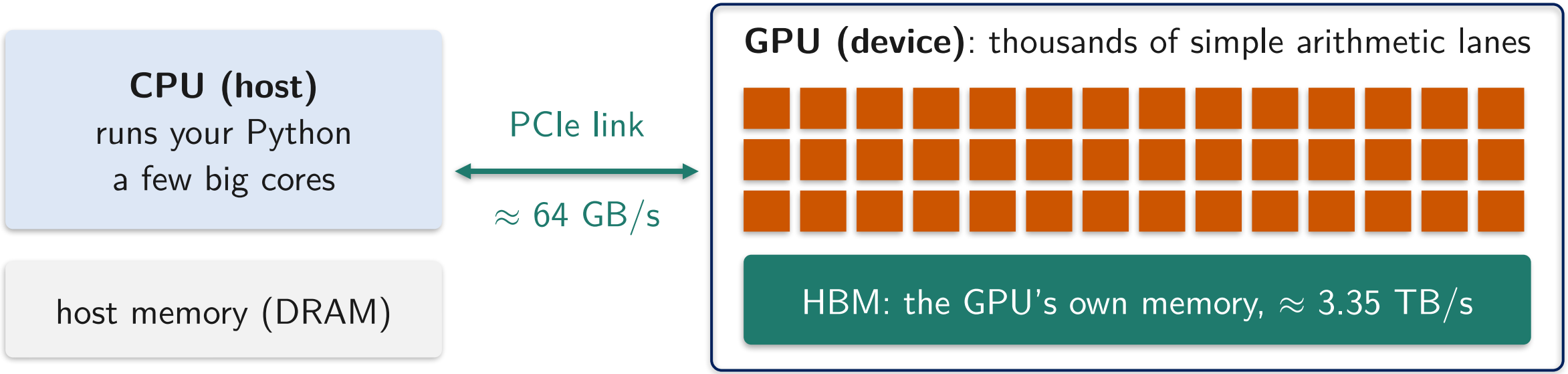
One **H100 GPU**, at its peak
$\approx$ **10 years**

Key idea

A GPU is **thousands of times** faster than a CPU core at this work — and even one GPU is too slow. So: GPUs today, many GPUs in T2 and T3.

Source: Brown et al., NeurIPS 2020 (GPT-3 compute); CPU: assumed 3 GHz core, 2 AVX-512 FMA units; H100 datasheet (SXM5), BF16 peak.

What a GPU is, in plain words



1. Copy the data to HBM once, and keep it there.
2. Launch kernels: "run $z = x + y$ on all 10^8 elements."

Key idea

A **kernel**: one function the GPU runs on many elements at once. The CPU **launches** it.

Source: H100 datasheet (SXM5), HBM bandwidth; PCIe 5.0 x16, per direction.

Where GPUs show up

- **Graphics first:** every pixel is independent, so the chip is many simple cores.
- **2007, CUDA:** those cores become programmable for any arithmetic.
- **2012, AlexNet:** trained on two GPUs. Llama 3 used up to **16,384** H100s.
- **In PyTorch:** `x.to("cuda")`; every op on x then runs as GPU kernels.

The price

Fast only with **lots of independent work**

Data must fit in **80 GB** of GPU memory

Each launch costs CPU time: tiny jobs are **overhead-bound**

In our pipeline

Why GPUs? Training is a huge number of independent multiply-adds, and a GPU runs tens of thousands at once. In T1 and Lab 3 your kernels run on one.

Source: Krizhevsky et al., NeurIPS 2012; Llama Team, "The Llama 3 Herd of Models," 2024; H100 datasheet (SXM5).

Learning goals for today

- Explain why a GPU trades **latency** for **throughput**.
- Map **thread** → **warp** → **block** → **grid** onto the GPU's **SMs**.
- Name the memory levels — **registers, shared memory, L2, HBM** — with rough sizes and speeds.
- Compute a kernel's **arithmetic intensity** from FLOPs and bytes.
- Use the **roofline model** to predict compute- versus memory-bound, and estimate runtime **before** measuring.

A puzzle to start

Kernel A — vector add

$z = x + y$, $n = 10^8$ floats

1 add per element

Kernel B — matrix multiply

$C = A \cdot B$, 4096×4096

8,192 FLOPs per output

On the same GPU, which kernel reaches a larger fraction of the advertised peak FLOP/s?

Key idea

Kernel B gets close to peak; **kernel A** reaches well under 1% of it. For A, arithmetic is not the bottleneck — **moving the data is**.

Counting work: FLOPs

- **FLOP** = one floating-point add or multiply; **FLOP/s** is a rate.
- Dot product of length K: K multiplies + (K − 1) adds ≈ **2K FLOPs**.
- C = A · B with A (M × K), B (K × N): M · N dot products.

$$F = 2 \cdot M \cdot N \cdot K$$

$$M = N = K = 4096$$

$$F \approx 1.37 \times 10^{11}\ \text{FLOPs}$$

Compute

At 989 TFLOP/s this takes ≈ **0.14 ms** — *if* arithmetic were the only cost.

Source: H100 datasheet (SXM5), BF16 tensor-core peak without sparsity.

Counting work for our own model

Step (batch B = 64)	**FLOPs**
Embedding lookup	≈ 0 (memory only)
Hidden: $(64\times128)\cdot(128\times64)$, tanh	≈ 1.05 M
Logits: $(64\times64)\cdot(64\times322)$	≈ 2.64 M
Softmax + cross-entropy	≈ 0.1 M
Backward $\approx 2\times$ matmuls	≈ 7.4 M
One training step	**$\approx$ 11 MFLOP**

ModelConfig: vocab 322, context 8, embedding 16, hidden 64

34,338 parameters $\approx$ 137 KB (FP32)

Check

At 67 TFLOP/s, 11 MFLOP takes $\approx$ **0.17 µs**. Hold that thought.

Part 1

From CPU to GPU

Why build a processor with thousands of slow, simple cores?

Latency versus throughput

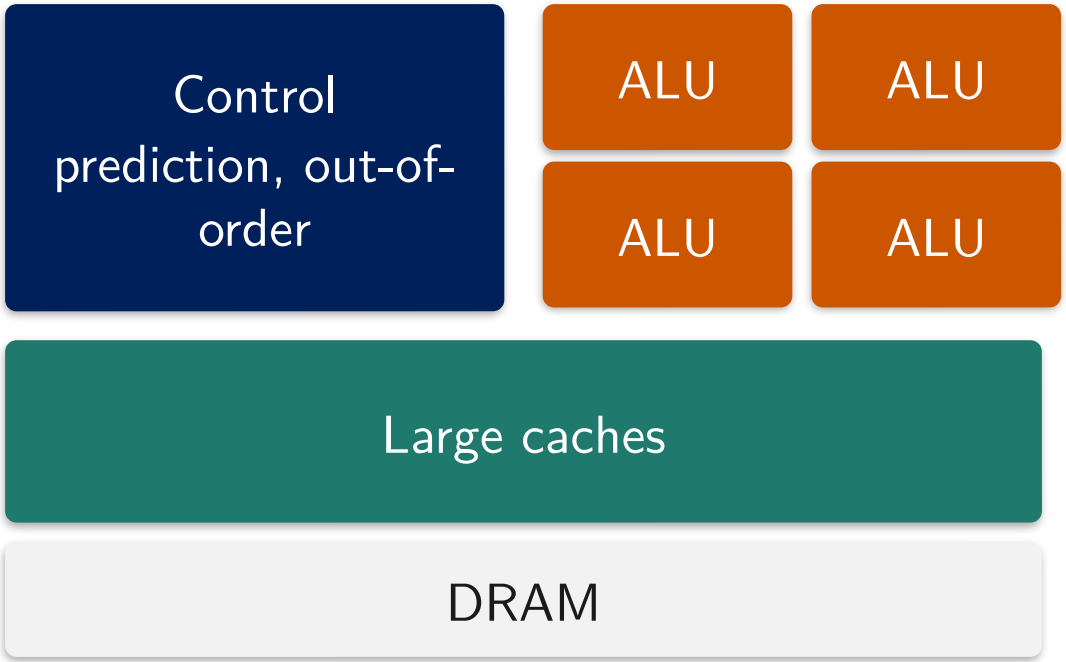
	Latency-oriented (CPU)	**Throughput-oriented (GPU)**
Goal	Finish **one** task fast	Finish **many** tasks per second
Cores	Tens of large cores	Thousands of simple lanes
Chip area	Caches, prediction, control	Arithmetic and registers
Hides slow memory by	Big caches, speculation	Switching to other threads

Key idea

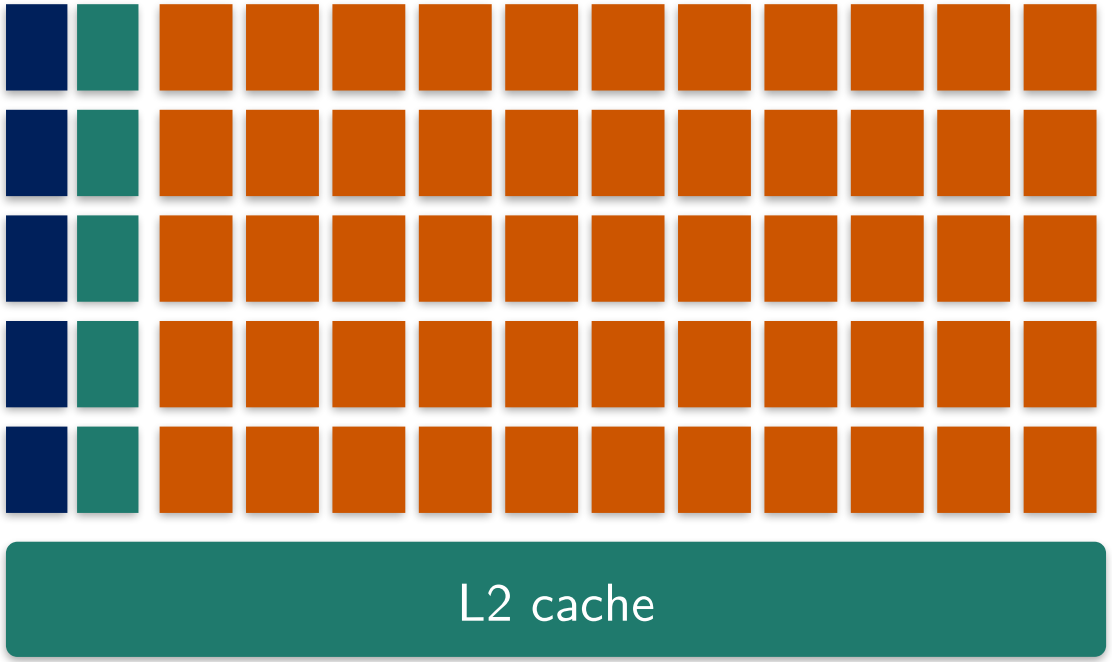
A sports car versus a fleet of buses. Training is a **throughput** problem.

Where the transistors go

CPU: a few powerful cores



GPU: many simple lanes

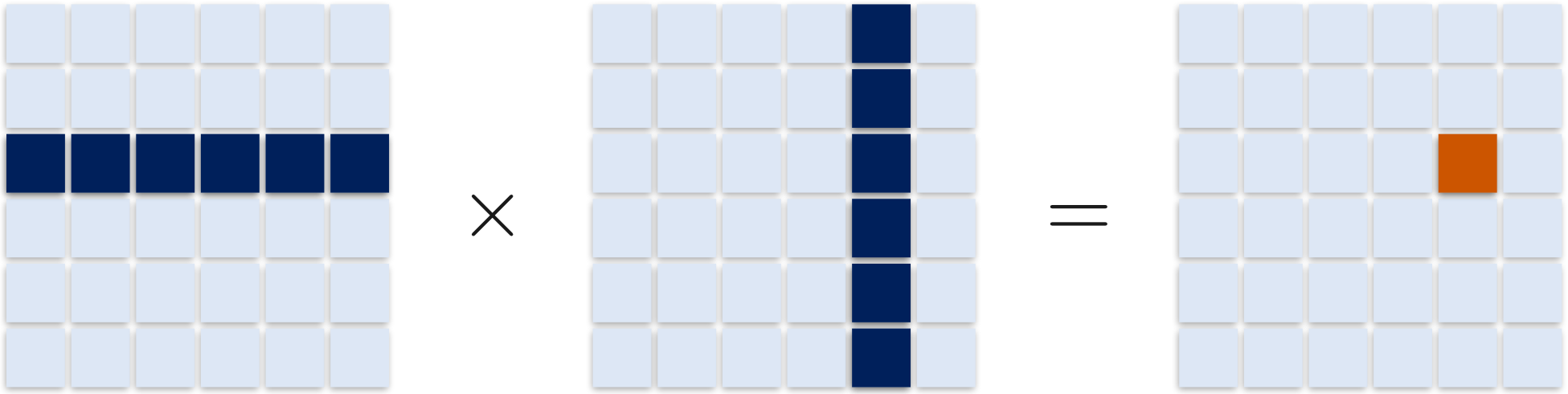


blue: control · green: cache · orange: arithmetic

Key idea

The GPU spends its area on arithmetic, so it needs **many threads** in flight to stay busy.

Why training fits the GPU



A row i × **B** column j = **C** C[i, j]

Key idea

Every C[i, j] is **independent**: $4096^2 \approx$ **16.8 million** tasks that can run at once.

Amdahl's law: the serial part limits you

$$\text{new time} = (1 - p) \cdot T + p \cdot T \,/\, s$$

$$\text{speedup} = 1 \,/\, (\,(1 - p) + p \,/\, s\,)$$

$$s \to \infty: \quad \text{speedup} \to 1 \,/\, (1 - p)$$

p = 0.50 $\to$ at most 2×

p = 0.90 $\to$ at most 10×

p = 0.95 $\to$ at most 20×

p = 0.99 $\to$ at most 100×

Watch out

If 5% of a training step is Python on the CPU, **no GPU** can make it more than 20× faster.

Check yourself

Two minutes with a neighbour.

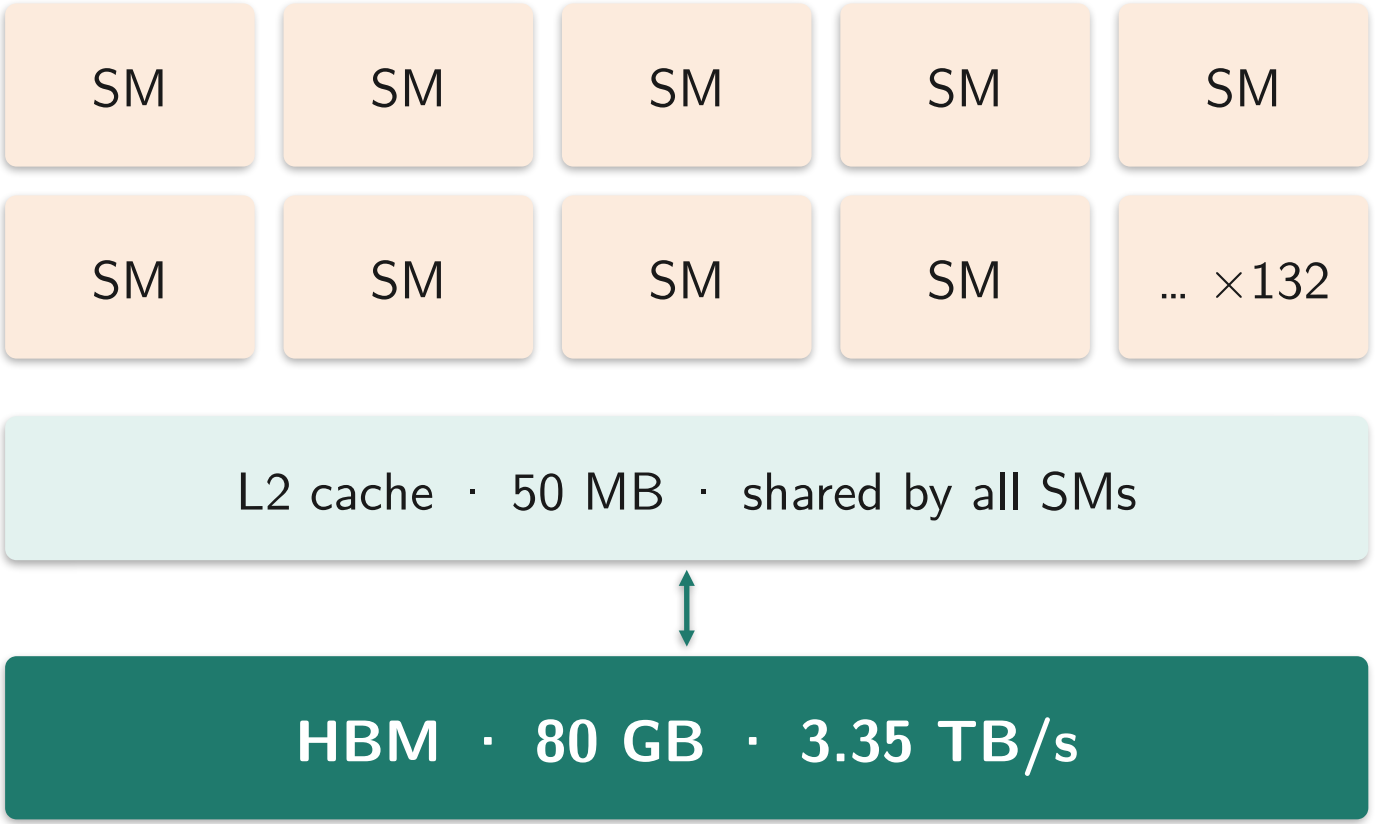
- Is training a latency problem or a throughput problem? Why?
- Name a computation that is hard to parallelize on a GPU, and say why.
- 80% of a program moves to a GPU that runs that part 100× faster. What is the overall speedup?

Part 2

The SIMT execution model

How does one program become 100,000 threads?

A GPU is a grid of streaming multiprocessors



$$\begin{aligned}
&\text{Peak FP32} \\
&= \text{SMs} \times \text{lanes} \times 2 \times \text{clock} \\
&= 132 \times 128 \times 2 \times 1.98\ \text{GHz} \\
&\approx 67\ \text{TFLOP/s}
\end{aligned}$$

A fused multiply-add $a \cdot b + c$ counts as 2 FLOPs.

Source: NVIDIA H100 Tensor Core GPU datasheet (SXM5); NVIDIA H100 Tensor Core GPU Architecture whitepaper (2022).

Inside one SM

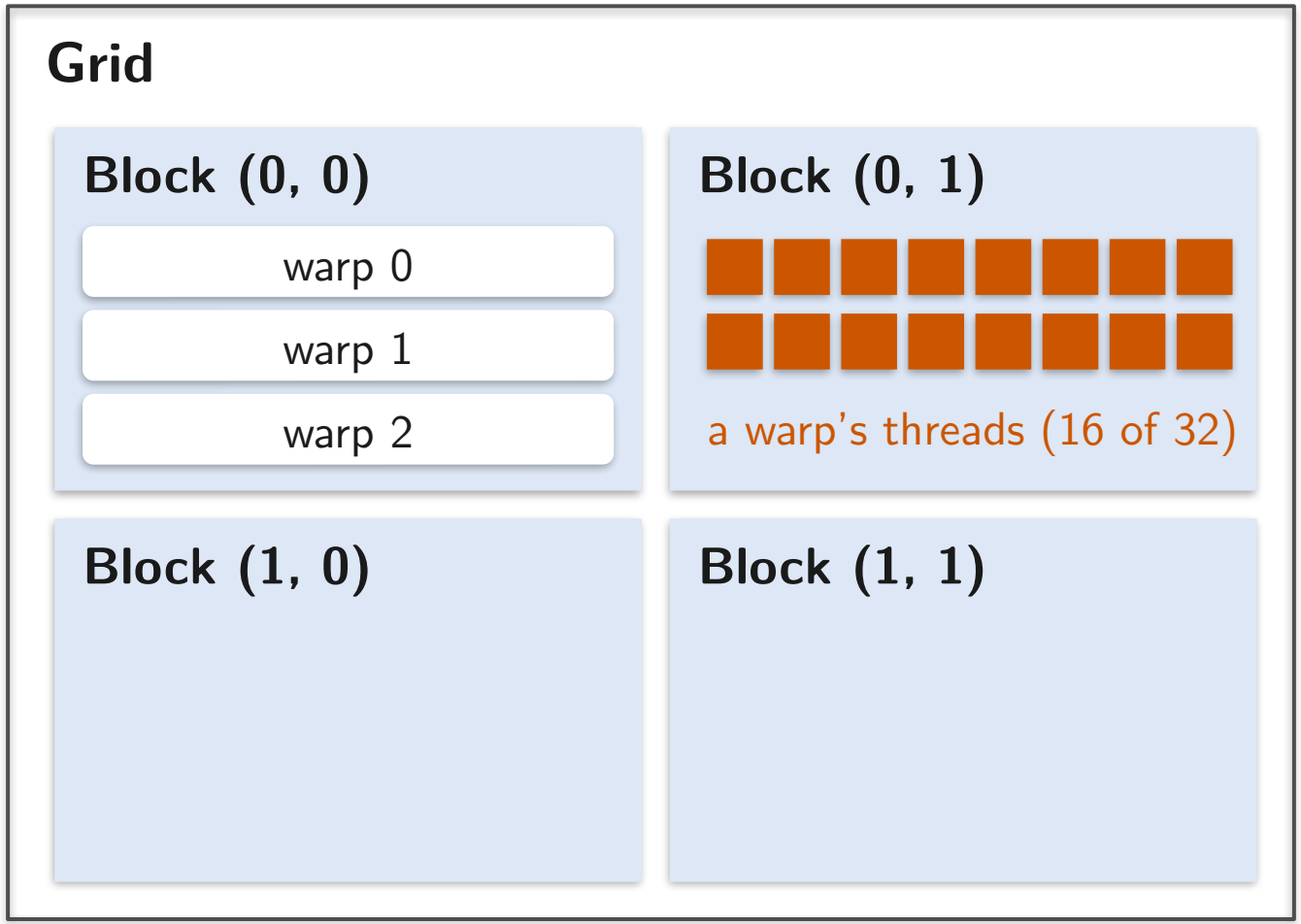
One of 4 sub-partitions

Warp scheduler	→ Picks a ready warp and issues **one instruction** each cycle.
Registers · 64 KB	→ Private, fastest storage: each thread has its own.
32 FP32 lanes	→ Execute that instruction for **32 threads at once**.
Tensor core	→ Multiplies small matrix tiles in one step (TF32, BF16, FP8, ...).

× 4 sub-partitions share 256 KB of L1 / shared memory (up to 228 KB shared)

Source: NVIDIA H100 Tensor Core GPU Architecture whitepaper (2022), SM description.

Software view: grid → block → warp → thread



Grid — every block of one launch; blocks must be **independent**.

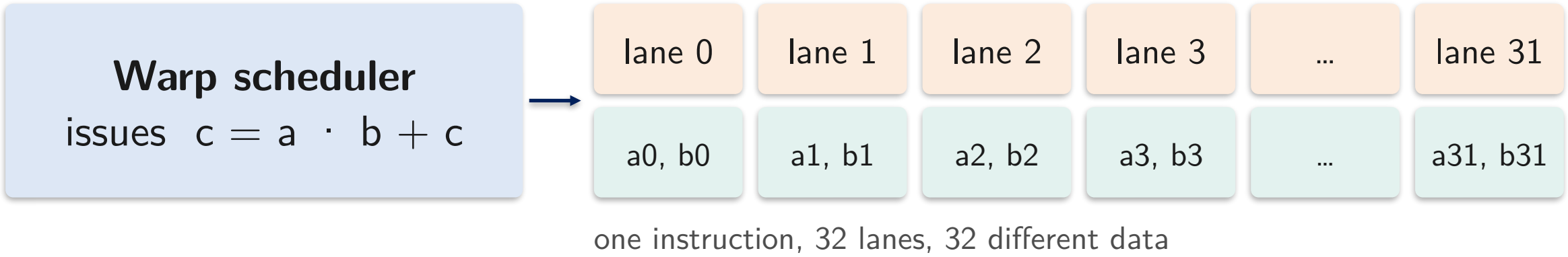
Block — up to 1024 threads on **one SM**; shares memory; can synchronize.

Warp — 32 threads that the hardware runs **together**.

Thread — one instance of the kernel, with its own registers.

Source: NVIDIA CUDA C++ Programming Guide: warp size 32; at most 1024 threads per block.

SIMT: single instruction, multiple threads



- One fetch and decode drives **32** operations $\rightarrow$ cheap control.
- Each thread has **its own registers** and index; code is written for one thread.
- A warp computes $(f(x_0), f(x_1), \ldots, f(x_{31}))$ with one instruction.

Branch divergence

```
i = thread_index()
if x[i] > 0:          # lanes 0-15
    y[i] = slow_a(x[i])
else:                 # lanes 16-31
    y[i] = slow_b(x[i])
```

$$\text{efficiency} = \text{active lanes} \;/\; 32$$

$$50/50 \text{ split:}$$

$$\text{time} \approx T(A) + T(B)$$

Pass 1: lanes run branch A



Pass 2: lanes run branch B

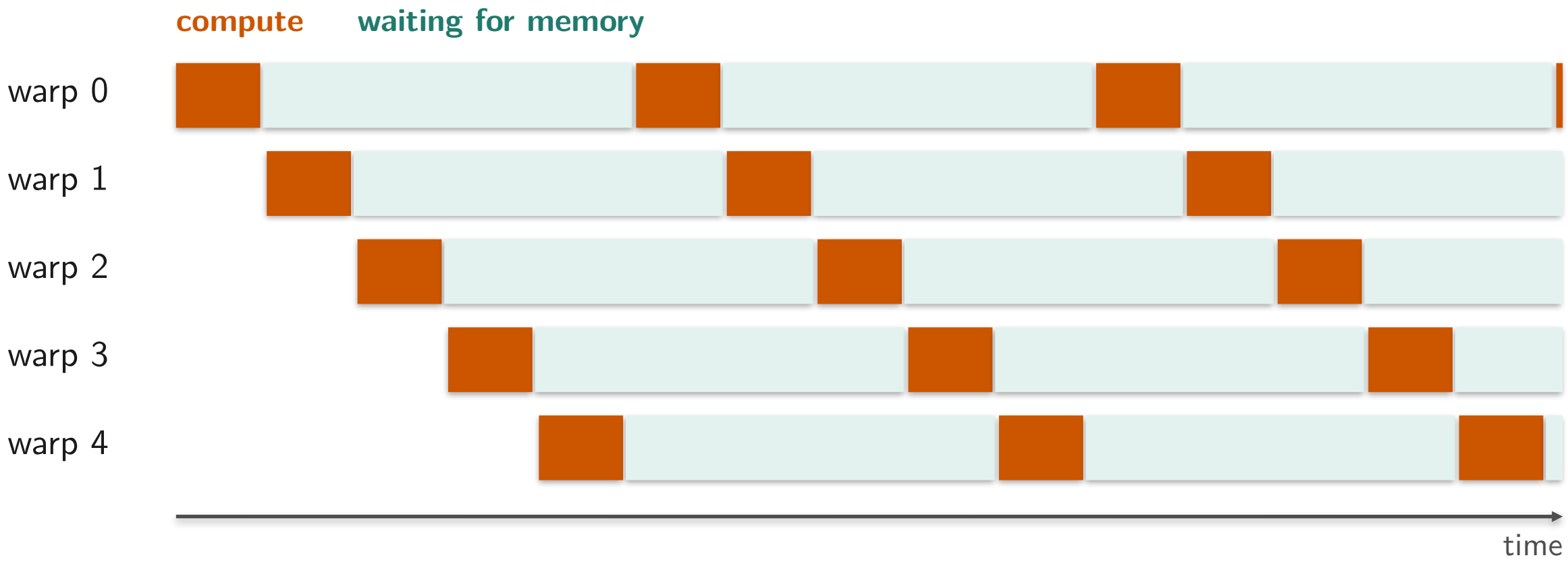


orange: active lane · grey: masked off

Watch out

Divergence costs time only **within a warp**; different warps may take different paths freely.

Hiding latency by switching warps



Key idea

While warp 0 waits, warps 1–4 compute; switching is free.

How many warps are enough? Little's law

$$\text{in flight} = \text{bandwidth} \times \text{latency}$$

$$= 3.35\ \text{TB/s} \times 600\ \text{ns} \approx 2\ \text{MB}$$

$$\text{per SM: } 2\ \text{MB} \,/\, 132 \approx 15\ \text{KB}$$

- 4-byte load per thread $\rightarrow \approx$ **3,800** threads per SM needed.
- An SM holds at most **2,048** threads $\rightarrow$ not enough.
- Fix: **16-byte** loads, several in flight per thread.

Key idea

Hiding latency needs **concurrency**: enough bytes in flight to cover the wait.

Source: Little (1961); H100 datasheet (bandwidth); latencies are rounded, from microbenchmarks (e.g. Luo et al., IPDPS 2024).

Occupancy: a worked example

Kernel: 256 threads per block · 128 registers per thread · 48 KB shared memory per block

- Registers: $65{,}536 \div 128 = 512$ threads $=$ **16 warps**
- Shared memory: $\lfloor 228\ \text{KB} \div 48\ \text{KB} \rfloor = 4$ blocks $\times$ 8 warps $=$ **32 warps**
- Thread limit: $2{,}048 \div 32 =$ **64 warps**
- Binding constraint: $\min(16, 32, 64) = 16$ warps $\rightarrow$ **25% occupancy**

Watch out

Occupancy helps **only** when the kernel is waiting on latency; it is a means, not a goal.

Source: NVIDIA CUDA C++ Programming Guide, technical specifications for compute capability 9.0 (H100).

Check yourself

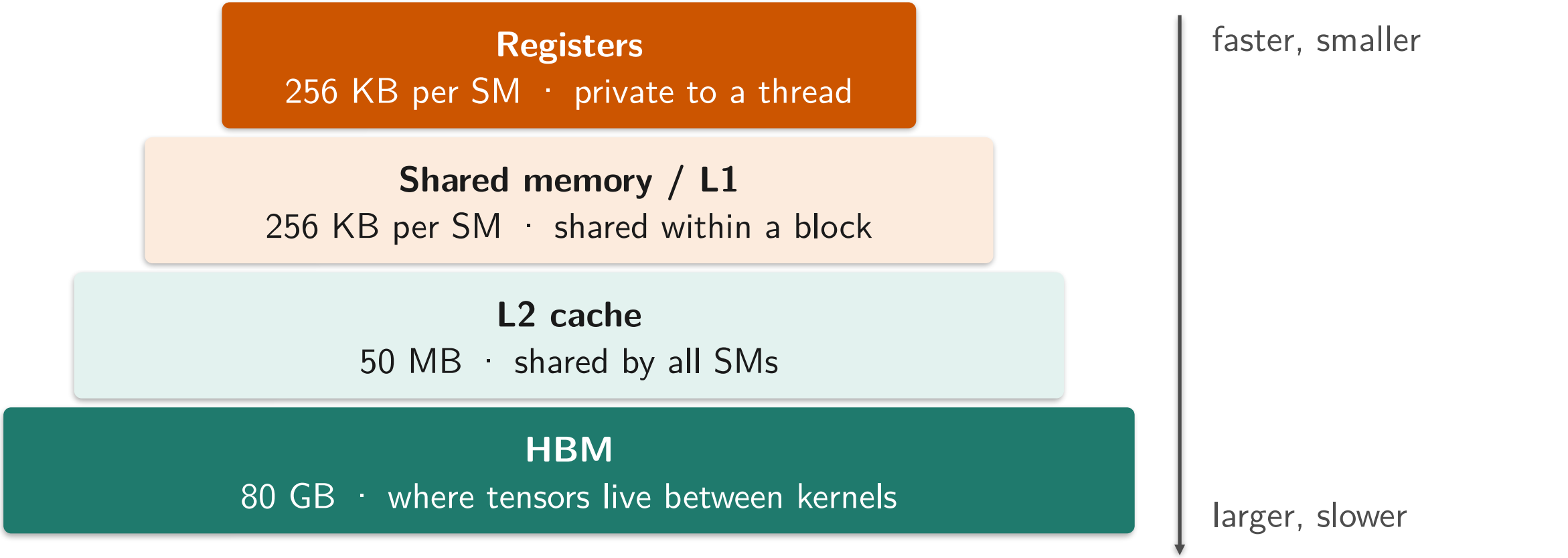
- How many threads form a warp? A block runs on how many SMs?
- `if i % 2 == 0: A() else: B()` — what does it cost, and how would you restructure it?
- Bandwidth 2 TB/s, latency 500 ns: how many bytes must be in flight?

Part 3

The memory hierarchy

Where does the data live, and how fast can we reach it?

The memory hierarchy



Key idea

Load from HBM **once**, reuse many times from registers or shared memory.

Source: NVIDIA H100 Tensor Core GPU datasheet (SXM5); NVIDIA H100 Tensor Core GPU Architecture whitepaper (2022).

The numbers to remember

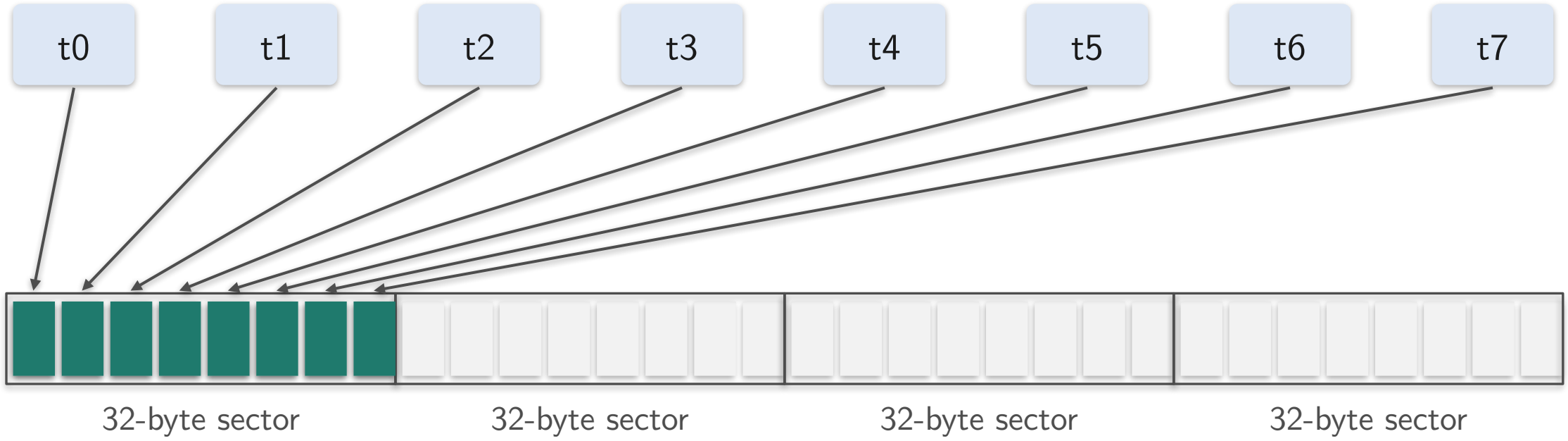
Level	Capacity	Latency (approx.)	Bandwidth (approx.)
Registers	256 KB / SM	$\approx$ 1 cycle	—
Shared memory / L1	256 KB / SM	$\approx$ 30 cycles	$\approx$ 33 TB/s (all SMs)
L2 cache	50 MB	$\approx$ 200–300 cycles	several TB/s
HBM	80 GB	$\approx$ 500–800 cycles	3.35 TB/s
Host RAM over PCIe 5	—	microseconds	$\approx$ 0.064 TB/s

Key idea

Host $\leftrightarrow$ GPU copies are $\approx$ **50×** slower than HBM: keep training data on the device.

Source: H100 whitepaper and datasheet; PCI-SIG PCIe 5.0; latencies are rounded, from microbenchmarks (e.g. Luo et al., IPDPS 2024).

Coalescing: neighbours read neighbours



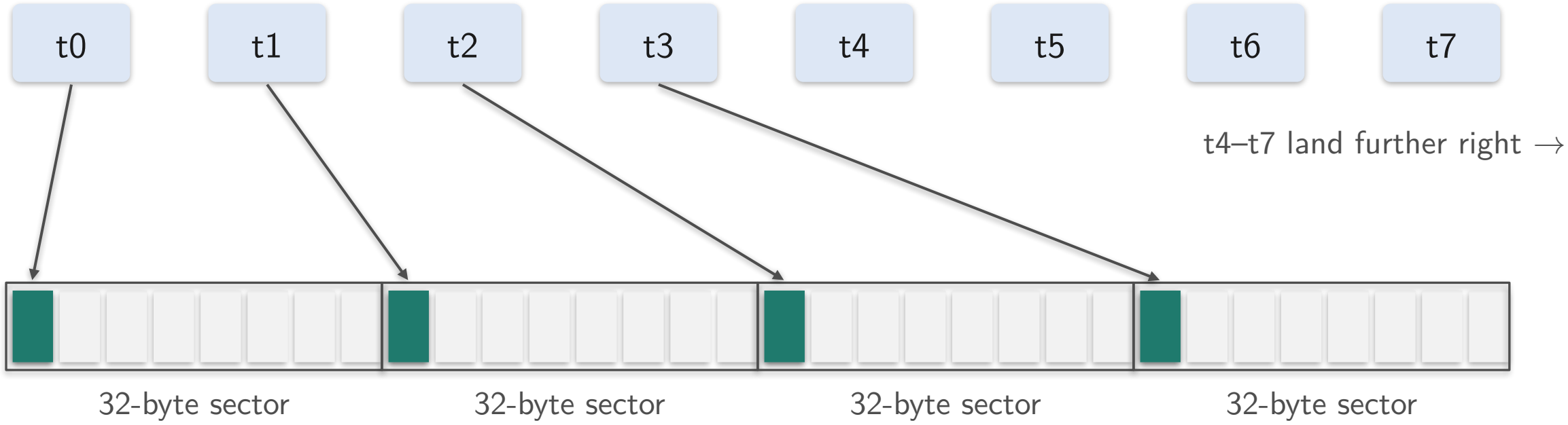
top: 8 of a warp's 32 threads · bottom: memory, one square per float

Memory

8 threads → one full sector; a warp of 32 → **128 bytes, all used**. Efficiency 100%.

Source: NVIDIA CUDA C++ Best Practices Guide, "Coalesced Access to Global Memory".

Strided access wastes bandwidth



top: 8 of a warp's 32 threads · bottom: memory, one square per float

Watch out

Each thread pulls a whole 32-byte sector for 4 useful bytes: 1,024 bytes moved, 128 used → **12.5%**.

Source: NVIDIA CUDA C++ Best Practices Guide, "Strided Accesses".

Row-major layout and strides

0,0	0,1	0,2	0,3	0,4	0,5
1,0	1,1	1,2	1,3	1,4	1,5
2,0	2,1	2,2	2,3	2,4	2,5
3,0	3,1	3,2	3,3	3,4	3,5

A: 4×6, row 1 highlighted



the same 24 numbers in memory: row 1 is contiguous

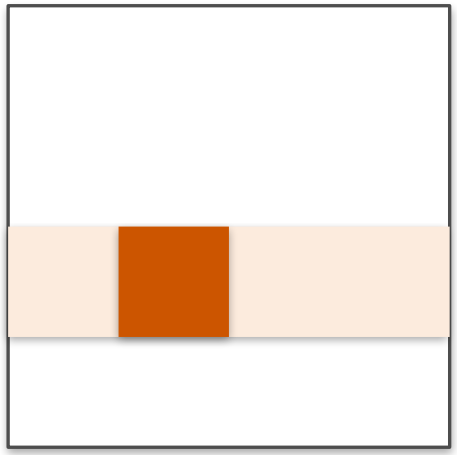
$$\text{offset}(i, j) = i \cdot N + j$$

$$\text{strides} = (N, 1)$$

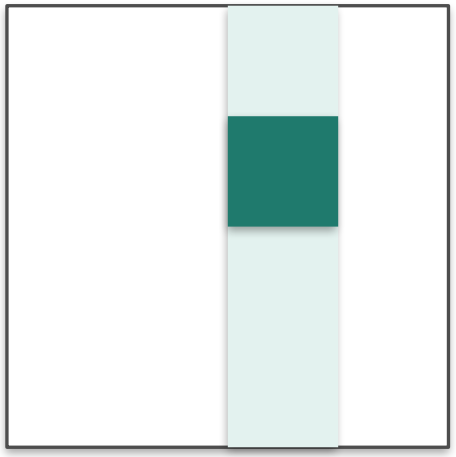
Key idea

A **row** has stride 1 (coalesced); a **column** has stride N (strided). `A.T` swaps strides, not data.

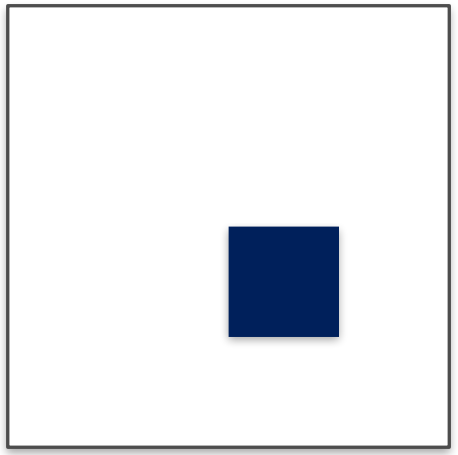
Tiling: reuse data on chip



A (M × K)



B (K × N)



C (M × N)

Naive: each output re-reads a whole row of A and column of B from HBM.

Tiled: load one T × T tile of A and of B into shared memory; use each value **T times**.

Key idea

HBM traffic falls by a factor of ≈ **T**: tiles move once, then are reused on chip.

Tiling, quantified

$$\text{naive:}\quad \text{reads} \approx 2\text{MNK} \;\Rightarrow\; I = 2\text{MNK} \,/\, 8\text{MNK} = 1/4$$

$$\text{tiled:}\quad \text{reads} \approx 2\text{MNK} \,/\, T \;\Rightarrow\; I \approx T \,/\, 4$$

$$T = 16 \to 4, \quad T = 32 \to 8, \quad T = 64 \to 16 \quad (\text{FLOP/byte})$$

Watch out

Bigger tiles mean more reuse, but more shared memory and registers → **lower occupancy**. You will tune T in T1.

Tensor cores and precision

Format	Bits	H100 peak (dense)	Typical use
FP32 (ordinary lanes)	32	$\approx$ 67 TFLOP/s	Reference math
TF32 (tensor cores)	19	$\approx$ 495 TFLOP/s	Faster FP32 matmuls
BF16 / FP16 (tensor cores)	16	$\approx$ 989 TFLOP/s	Training default
FP8 (tensor cores)	8	$\approx$ 1,979 TFLOP/s	Large-model training

Watch out

These peaks apply **only to matrix products**. Elementwise work (tanh, softmax, adds) runs on ordinary lanes and is usually memory-bound.

Source: H100 datasheet (SXM5); tensor-core values are the datasheet's sparse peaks divided by 2.

Check yourself

- Who manages shared memory? Who manages L2?
- 32 threads each read one float from a different **column** of a row-major matrix. What fraction of the bytes moved is used?
- What intensity does T = 32 tiling give in FP32? Why not simply use T = 128?

Part 4

The roofline model

Compute-bound or memory-bound? Predict before you measure.

Arithmetic intensity

$$I = F \,/\, Q$$

$$F = \text{FLOPs performed}, \quad Q = \text{bytes moved to/from HBM}$$

Kernel	F	Q (minimum)	I (FLOP/B)
Vector add z = x + y (FP32)	n	12n	$1/12 \approx 0.08$
Row softmax, B × V (FP32)	$\approx 5BV$	$8BV$	≈ 0.6
Matmul n × n (BF16, ideal reuse)	$2n^3$	$6n^2$	n / 3

Key idea

Q is the **minimum** traffic: every input read once, every output written once. Real kernels can only do worse.

Two lower bounds on runtime

$$T \geq F \,/\, \pi \qquad \text{(arithmetic)}$$

$$T \geq Q \,/\, \beta \qquad \text{(memory)}$$

$$T \geq \max(\, F \,/\, \pi \;,\; Q \,/\, \beta \,)$$

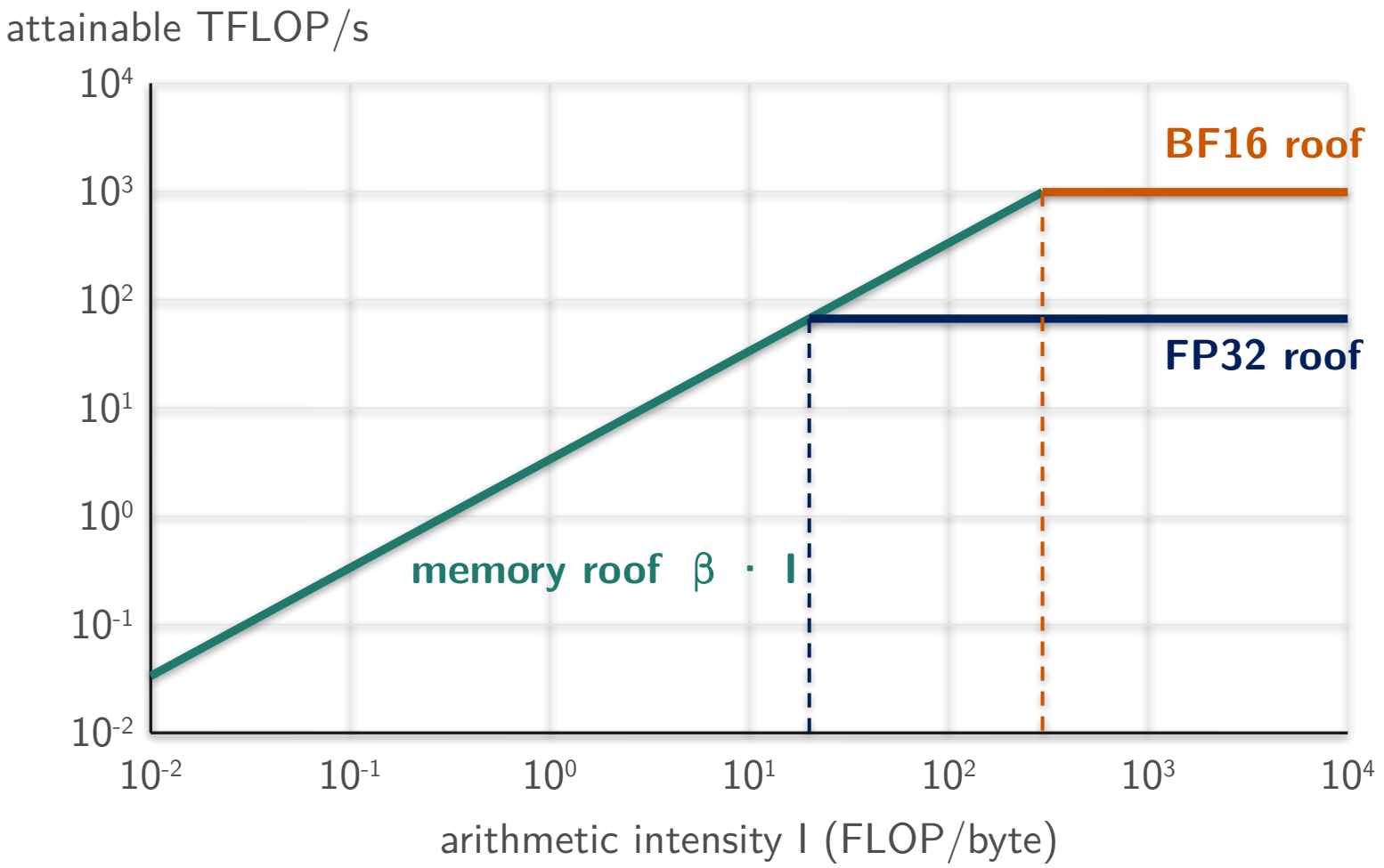
$$\text{attainable FLOP/s} = F \,/\, T \leq \min(\, \pi \;,\; \beta \cdot I \,)$$

Key idea

Ridge point $I^* = \pi \,/\, \beta$: below it, memory limits you; above it, arithmetic does. Assumes perfect overlap — an optimistic bound.

Source: Williams, Waterman & Patterson, "Roofline," Communications of the ACM 52(4), 2009; H100 datasheet (SXM5).

The roofline for an H100



FP32 roof: 67 TFLOP/s

BF16 roof: 989 TFLOP/s

ridges $\approx$ 20 and 295

$\leftarrow$ memory-bound: move fewer bytes

compute-bound $\rightarrow$ use faster arithmetic

Source: Williams, Waterman & Patterson, "Roofline," Communications of the ACM 52(4), 2009; H100 datasheet (SXM5).

Worked example 1: vector add

- $n = 10^8$ floats (FP32); $F = 10^8$ FLOPs
- $Q = 3 \times 4 \times 10^8 = 1.2 \times 10^9$ bytes
- $I = F / Q = 1/12 \approx 0.08$ FLOP/B $(\ll 20)$
- $F / \pi = 10^8 / 67 \times 10^{12} \approx 1.5$ µs
- $Q / \beta = 1.2 \times 10^9 / 3.35 \times 10^{12} \approx$ **0.36 ms**

Verdict

Memory-bound by $\approx 240\times$. A measured time near 0.36 ms means the kernel is already **optimal**: it runs at the memory roof.

Worked example 2: matrix multiply

$$F = 2n^3, \quad Q = 6n^2, \quad I = n \,/\, 3$$

$$\text{compute-bound when } n \,/\, 3 > 295 \;\Leftrightarrow\; n \gtrsim 886$$

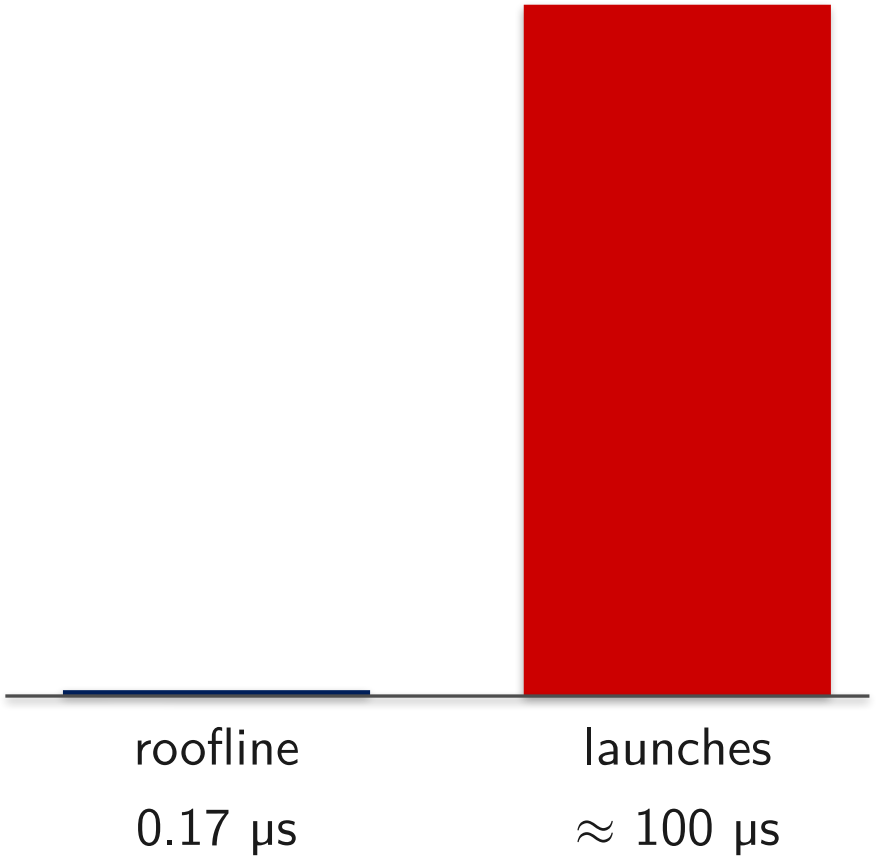
n	I (FLOP/B)	Bound	Predicted time
256	85	memory (on paper)	0.12 µs → launch overhead dominates
1024	341	compute (just)	≈ 2.2 µs
4096	1365	compute	≈ 0.14 ms

Watch out

Tiny problems hit a third limit: **fixed overhead** — a kernel launch costs several µs.

Worked example 3: our training step

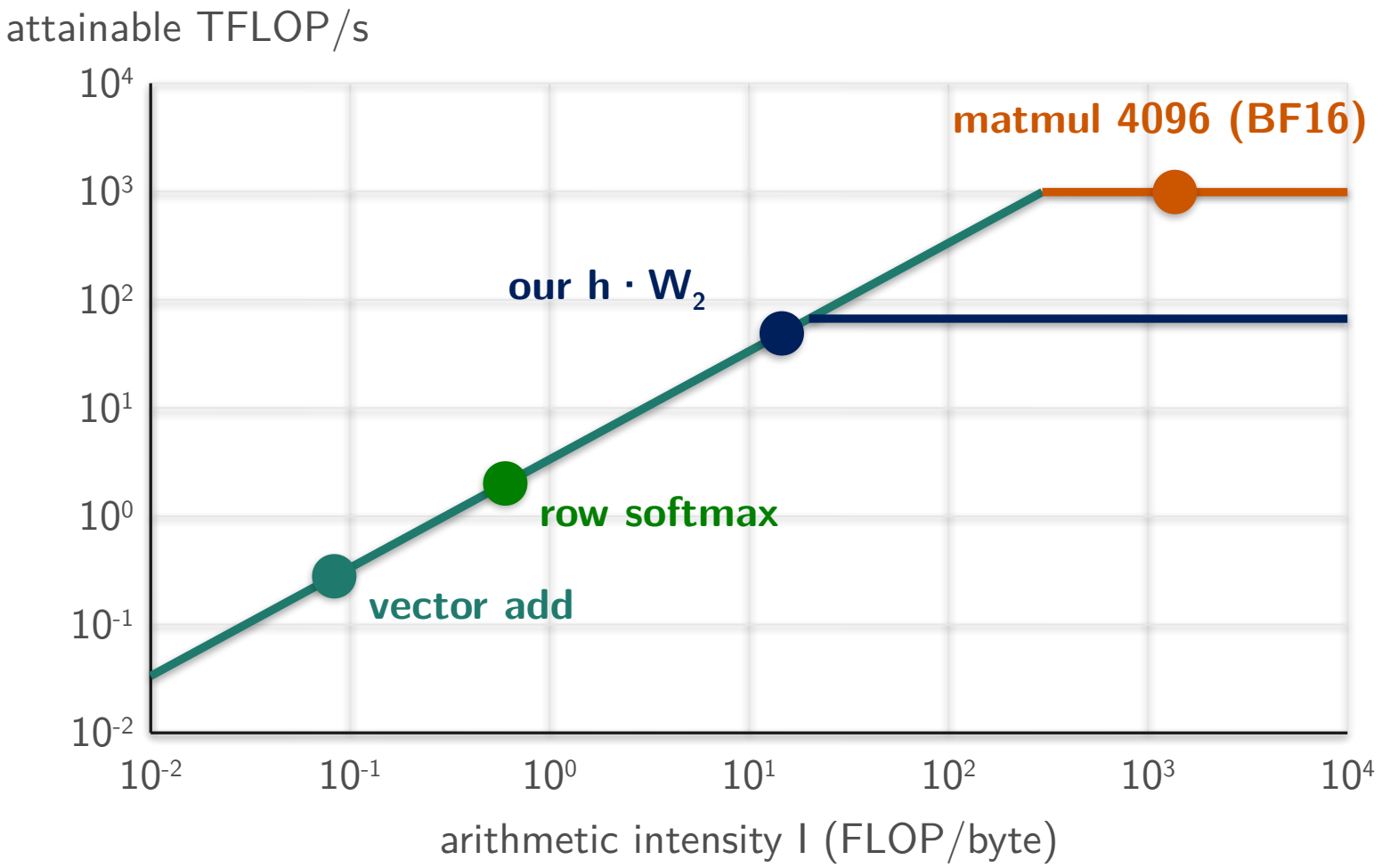
- Logits $h \cdot W_2$: $(64 \times 64) \cdot (64 \times 322)$, FP32
- $F \approx 2.64$ M FLOPs; $Q \approx 181$ KB; $I \approx 14.6$
- Roofline time ≈ 0.05 µs; whole step $\approx$ **0.17 µs**
- ≈ 25 kernel launches $\times \approx 4$ µs $\approx$ **100 µs**



In our pipeline

Overhead-bound: launches cost $\approx 600\times$ the roofline time. Fix: **fuse**, bigger batches, **compile**.

Placing kernels on the roofline



- **Tensor cores** help only the large matmul.
- **Fusion** helps the left side: it cuts Q.

Source: Williams, Waterman & Patterson, "Roofline," Communications of the ACM 52(4), 2009; H100 datasheet (SXM5).

Fusion: fewer bytes, same FLOPs

Unfused: 3 kernels

$t_1 = a \cdot x$ read x, write t_1

$t_2 = t_1 + b$ read t_1, write t_2

$y = \tanh(t_2)$ read t_2, write y

Q = 24n bytes

Fused: 1 kernel

$y = \tanh(a \cdot x + b)$

read x once, write y once

t_1, t_2 stay in registers

Q = 8n bytes

$$\text{same } F;\ \ Q\text{: } 24n \to 8n;\ \ \text{memory-bound} \Rightarrow \text{predicted speedup} \approx 3\times$$

In our pipeline

T1: write one Triton **reduction or fused elementwise** kernel, and predict its speedup **before** timing it.

What the roofline leaves out

Effect	Symptom	Covered in
Launch / Python overhead	Small problems far below both roofs	Today; Week 9
Uncoalesced access	Q far above the minimum	Today; Lab 3
Low occupancy	Latency not hidden; below the roof	Today; T1
Branch divergence	Idle lanes; below the compute roof	Today
Host $\leftrightarrow$ device copies	GPU idle between kernels	Week 10

Key idea

When measurement and prediction disagree, the **gap is the diagnosis**.

IN-CLASS BUILD

Predict a kernel bottleneck

Work in pairs. Predict first; measure in Lab 3.

Exercise: predict three kernels

Kernel (FP32)	F	Q	I	Bound?	Time
1. ReLU $y = \max(x, 0)$, $n = 10^8$					
2. Row sums of a 4096×4096 matrix					
3. Matmul 2048×2048, tiles $T = 64$					

USE

$\pi = 67$ TFLOP/s, $\beta = 3.35$ TB/s

One comparison = one FLOP.

STRETCH

For kernel 3, compare the tiled estimate with the ideal-reuse estimate.

Exercise: solutions

Kernel (FP32)	F	Q	I	Bound	Time
1. ReLU, $n = 10^8$	10^8	8×10^8 B	0.125	memory	$\approx$ 0.24 ms
2. Row sums, 4096^2	1.7×10^7	6.7×10^7 B	0.25	memory	$\approx$ 20 µs
3a. Matmul, $T = 64$	1.7×10^{10}	F / 16	16	memory	$\approx$ 0.32 ms
3b. Matmul, ideal reuse	1.7×10^{10}	5.0×10^7 B	341	compute	$\approx$ 0.26 ms

Key idea

3a and 3b **bracket** reality. In Lab 3 you will measure kernel 1 and see how close you get to 0.24 ms.

Measuring fairly

```
def bench(fn, repeats=20):
    fn(); torch.cuda.synchronize()    # warm up
    times = []
    for _ in range(repeats):
        t0 = time.perf_counter()
        fn()
        torch.cuda.synchronize()
        times.append(time.perf_counter() - t0)
    return statistics.median(times)

x = torch.randn(10**8, device="cuda")
t = bench(lambda: torch.relu(x))
print(8 * x.numel() / t / 1e9, "GB/s")
```

- **Synchronize**: launches are asynchronous.
- **Warm up**: the first call sets up.
- **Repeat**: report median and spread.
- **Convert** to GB/s and compare with the roof.

In our pipeline

T1 checkpoint: a correct kernel **and** a correct timing harness.

Key takeaways

- GPUs trade single-thread **latency** for **throughput**; training is a throughput problem.
- Threads run in **warps of 32** (SIMT); divergence within a warp wastes lanes.
- GPUs hide memory latency by **switching among many warps** (Little's law, occupancy).
- **Coalescing** and **tiling** reduce the bytes that reach HBM.
- Roofline: $\mathbf{T \geq \max(F/\pi,\ Q/\beta)}$ — compute $I = F/Q$, compare with π/β, predict before measuring.
- Our small MLP is **overhead-bound** — the motivation for fusion and compilation.

Next time, and readings

When	What
Lecture 2 (this week)	GPU programming model: blocks, grids, masks, reductions in Triton
This week	Midterm (data systems), evening, time TBA
After Lecture 2	T1 and Lab 3 released
Week 9	Frameworks: tensors, dispatch, memory; eager vs compiled

READ BEFORE LECTURE 2

CUDA C++ Programming Guide: *Programming Model* and *Hardware Implementation*.
Williams, Waterman & Patterson, "Roofline," CACM 2009.